

AYUSHI PATEL

AI-Driven Data Analyst | Applied Data Science | Machine Learning

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PROFESSIONAL SUMMARY

AI-driven Data Analyst with 4+ years of experience translating complex datasets into measurable business impact across financial risk, retail, and industrial domains. Delivered a 25% improvement in decision-making speed, 40% reduction in data processing time, and 20% boost in predictive model accuracy by combining SQL, Python, and machine learning with modern AI tooling. Proven ability to architect end-to-end analytics pipelines, from raw data ingestion and ETL automation to executive-level dashboards and LLM-powered reporting. Hands-on experience with Generative AI workflows (GPT, LangChain, FAISS, RAG pipelines) and Explainable AI (SHAP) to build trustworthy, interpretable models. Adept at working cross-functionally with VP-level stakeholders to define KPIs, align analytics strategy with business goals, and drive revenue growth through data.

TECHNICAL SKILLS

Programming & Data

Python (Pandas, NumPy, Scikit-learn)

SQL (Snowflake, BigQuery, PostgreSQL)

R

AI & Machine Learning

Regression & Classification

Clustering

Time Series Forecasting

Anomaly Detection

XGBoost & Random Forest

SHAP (Explainable AI)

A/B Testing

Generative AI & LLM

GPT-4

Prompt Engineering

LangChain

FAISS

Hugging Face

RAG Pipelines

Vector Databases

NLP

Data Visualization & BI

Power BI

Tableau

Looker

Sigma

Plotly

Streamlit

Advanced Excel

Data Engineering, MLOps & Cloud

ETL Pipelines

DBT

Apache Airflow

MLflow

Git & Docker

AWS (S3, Redshift)

GCP (BigQuery)

Snowflake

Databricks

PROFESSIONAL EXPERIENCE

Data Analyst

Mar 2024 - Present

ISquareInfo LLC | USA

- Architected 10+ interactive Power BI and Tableau dashboards integrated with SQL, Python, and Snowflake, cutting time-to-insight by 25% for senior leadership and enabling real-time market and risk monitoring across 5 business units.
- Engineered automated ETL pipelines on BigQuery using Python and Apache Airflow, reducing data processing time by 40% and eliminating 15+ hours of weekly manual data preparation work.
- Built and deployed XGBoost and Scikit-learn forecasting models for business planning, improving predictive accuracy by 20% and supporting \$2M+ in quarterly revenue forecasting decisions.
- Integrated GPT-4-based LLM workflows with prompt engineering and LangChain to automate insight generation from structured datasets, reducing manual reporting effort by 30% and enabling analysts to focus on higher-value tasks.
- Optimized regulatory reporting pipelines in Power BI and Excel, reducing manual compliance effort by 30%, strengthening audit readiness, and ensuring 100% on-time delivery of finance and governance reports.
- Partnered with VP-level leaders to define KPI frameworks across 3 departments, redesigning reporting structures to align analytics output with corporate OKRs and increase executive visibility.
- Conducted A/B testing and cohort analysis to evaluate campaign performance and product changes, surfacing insights that contributed to a 12% improvement in customer engagement metrics.

Business Data Analyst

Dec 2019 - Aug 2021

Systech Solutions | India

- Transformed 500K+ rows of PLC/SCADA industrial sensor data into actionable intelligence using Python, SQL, and Databricks, boosting operational efficiency by 35% and enabling faster data-driven production decisions.
- Built organization-wide KPI dashboards using Power BI and Tableau with DBT and Fivetran integration, standardizing performance tracking across 8 business units and reducing reporting lag from 3 days to same-day.
- Deployed anomaly detection and trend-forecasting models using Scikit-learn on live sensor streams, predicting equipment failures 48 hours in advance and reducing unplanned downtime incidents by 20%.
- Automated recurring data quality checks and validation scripts in Python, reducing data errors by 25% and improving confidence in downstream analytics and reporting.
- Partnered with operations and leadership teams to design KPI frameworks and standardized reporting templates, expanding executive visibility into plant-level performance and supporting strategic capacity planning.
- Collaborated with cross-functional teams to document data definitions, lineage, and business rules, establishing a shared data dictionary used across 4 departments.

PROJECTS

Rag-Powered Analytics Assistant (Generative AI + LLM)

- Designed and built a production-style GenAI analytics assistant using LangChain, FAISS, GPT-4, and Python, enabling natural-language Q&A over structured retail business datasets (500K+ records across sales, inventory, and customer data).
- Engineered a full end-to-end RAG pipeline: document ingestion and chunking, vector embedding via Hugging Face sentence-transformers, FAISS-based semantic retrieval, and GPT-4 response generation with source attribution.
- Applied advanced prompt engineering patterns (chain-of-thought, few-shot, system personas) to generate executive-ready trend summaries, anomaly alerts, and business recommendations from raw data automatically.
- Replaced approximately 50% of manual SQL reporting tasks (estimated 10+ hours/week saved), deploying the assistant as an interactive Streamlit app with session-based conversation history.
- Evaluated retrieval quality using precision@k and response quality using ROUGE-L scoring, iterating on chunk size and overlap parameters to achieve optimal retrieval accuracy.

SmartRetail Insights - End-to-End AI Analytics System

- Developed a full analytics ecosystem using Python, SQL, Power BI, and MLflow to unify sales forecasting, customer segmentation, and revenue performance tracking for a simulated retail environment with 1M+ transaction records.
- Improved marketing ROI by 15% through XGBoost-based propensity models and RFM customer segmentation, identifying high-value customer clusters and enabling targeted campaign allocation.
- Built automated data pipelines with Pandas and SQLAlchemy to ingest, clean, and transform raw POS and web data, reducing data preparation time by 35% compared to manual methods.
- Tracked all model experiments, parameters, and metrics with MLflow, enabling reproducible model comparisons and seamless promotion of the best-performing model to the reporting layer.
- Delivered an executive Power BI dashboard with drill-down by region, product category, and customer segment, enabling non-technical stakeholders to self-serve insights without analyst involvement.

EDUCATION

Master of Science, Information Systems

Jan 2022 - Dec 2023

Northeastern University | Boston, MA

Bachelor of Technology, Computer Engineering

Aug 2017 - Jul 2021

Cummins College of Engineering for Women | India